

development LADAKH

■ By Ramesh Vinayak in Leh

At 18,380 ft in the Himalayas, Khardung La, a wind-swept pass with scanty oxygen on the world's highest motorable road in Ladakh, is the ultimate milestone for record-crazy adventure seekers. But, Ritesh Arya's fascination with craggy and barren mountains runs deeper, literally. This intrepid hydrogeologist is about to succeed in his quest for ground water on Khardung La—a feat that could surpass his own world record of digging borewells at high altitudes. Already, sparkling clear ground water is streaming out of two borewells he had dug recently at South Pullu and North Pullu, army posts and snow shelters on either side of the pass at 15,300 ft and 15,400 ft, respectively.

Until a month ago, the only source of drinking water here were water tankers from distant Leh and Partapur at the base of the Siachen glacier. "It's nothing short of a miracle to get ground water at this height," gushes a Junior Commissioned Officer of the military police post at South Pullu, an area where granite rocks abound which, according to conventional geology, are too impervious to hold any ground water. But Arya perceived a narrow valley of rock debris at the base of the receding Khardung glacier, now 6 km from the road, as the most definite indicator of ground water charged by the melting glacier. And, two days after a rig drilled a 300-ft deep hole, it struck a ground water reservoir.

Arya plans his drilling operations after studying the exposed rock faces in the landscape. Such exploits come naturally to him, a diminutive 39-year-old who holds a PhD degree in geology. By combining his hands-on expertise in Himalayan geology with an unconventional approach, this hydrogeologist-turned-professional driller has broken new ground on scientific exploitation of ground water in the high-altitude, cold

desert of Ladakh. In the past 12 years, Arya has dug more than a hundred borewells in inhospitable and treacherous terrains where no geologist or government agency has ventured before. From Siachen glacier to the China border, the Indus plains of Leh and the Kargil heights, his explorations have ensured all-weather ground water supplies to the army and civilians alike.

More significantly, Arya's pioneering research is likely to redefine Himalayan hydrology and change the traditional schemes for drinking water and irrigation in rain-deficit Ladakh, which has so far been harnessing

mostly surface water from the river Indus or glacier-fed streams. Apart from perennial shortage, there is also the problem of silt in glacier melt in summer and freezing of surface water sources in winter. In Leh town, for example, only 10 per cent of the population—which rises from 15,000 to 50,000 every summer due to tourist inflow—has access to ground water through public taps, the rest depends on water tankers. Ground water exploitation in Ladakh, undertaken by the Central Ground Water Board (CGWB), has been confined to areas along the Indus river. Exploration beyond that

THE INCREDIBLE W

An intrepid hydrogeologist creates a record by digging for water in the world's highest altitudes



"It's like rediscovering the principles of geology and physics operating in high-altitudes."

RITESH ARYA, HYDROGEOLOGIST